

Option D. Student thought that there is no deformation in the wire.

9 Ans: B

By drawing the free body diagram of the car for each bridge, realise that for convex bridges

Weight of car – Normal contact force on car by road = mv^2/r

Normal contact force on car by road = Weight of car – mv^2/r (smaller than weight)

The convex bridge which has the shorter radius will exert the least normal contact force on the car.

By Newton's 3rd Law, force exerted by car has same magnitude as the normal contact force

10 Ans: B